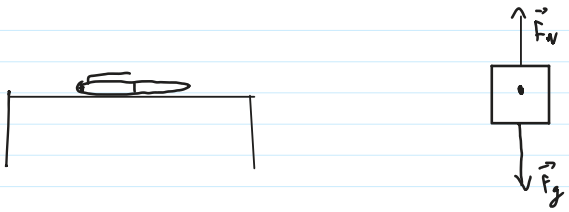
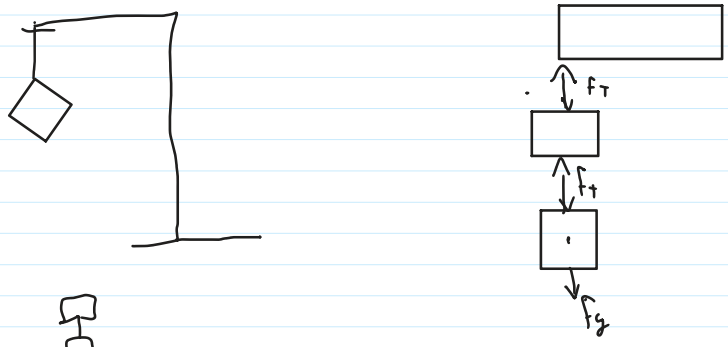
Practice

1. Draw a simple system diagram and an FBD for each of the following objects. K/U C A
   (a) a pen sitting on a table
   (b) a rope connected to a crane raising a piano vertically upward at a constant speed
   (c) a lamp that has just begun falling from a table to the floor; air resistance is negligible
   (d) a dresser that is being pulled to the right up a ramp into a delivery truck by a cable parallel to the ramp; the ramp is at an angle of 14° above the horizontal

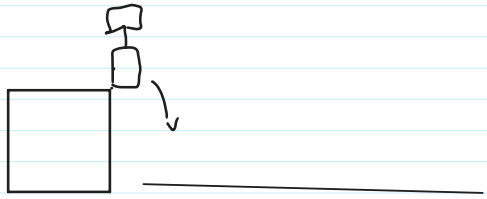
1)



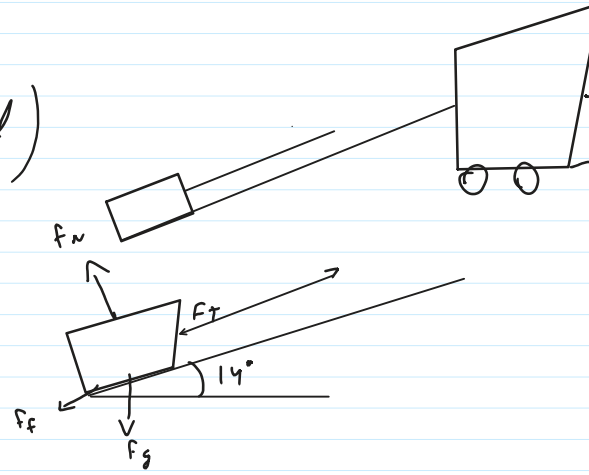
b)



c)

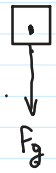


d)



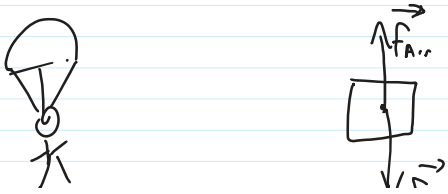
2. You throw a ball vertically upward. Air resistance is negligible. Draw an FBD of the ball
   (a) just after it leaves your hand
   (b) at the top of its motion
   (c) as it is falling back down K/U C A

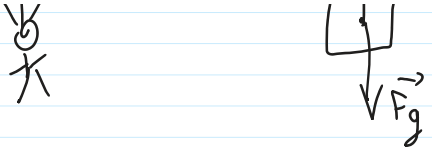
a) b) c)



3)

3. A skydiver whose parachute is open can see his instantaneous height above ground level on an electronic screen. The skydiver has reached terminal speed. (Recall from earlier studies that when an object falls at terminal speed it is falling at a constant velocity.) For this question, assume the skydiver and the parachute together act as one body. Draw a system diagram and an FBD for the situation. K/U C A





1. Determine the net force acting on each of the following objects. In each case assume that all forces acting on the object are given. K/U T/I
   (a) At an instant when a soccer ball is slightly off the ground, a player kicks it, exerting a force of 25 N at 40.0° above the horizontal. The force of gravity acting on the ball is 4.2 N [down]. [ans: 23 N [32° above the horizontal]]
   (b) Two children pull a sled across the ice. One child pulls with a force of 15 N [N 35° E], and the other pulls with a force of 25 N [N 54° W]. There is negligible friction acting on the sled. [ans: 29 N [N 23° W]]
   (c) In a circus act, a performer with a force of gravity on her of 4.4×10^2 N is lifted by two different ropes at the same time. One rope exerts a tension of 4.3×10^2 N [up and 35° left], and the other performer exerts a force of 2.8×10^2 N [up]. [ans: 3.1×10^2 N [up 52° left]]
2. Two tractors pull a large rock east through a construction site with a net force of zero on the rock. Tractor 1 exerts a force of 1.2×10^4 N [E 12° N] on the rock, and tractor 2 exerts a force of 1.2×10^4 N [E 12° S]. K/U T/I
   (a) Calculate the force of friction acting on the rock. [ans: 2.3×10^4 N [W]]
   (b) Discuss two ways someone could spot that the force of friction on the rock must be to the west before solving the problem.

$\sum \vec{F}$ = Net force

1a)

$\vec{F}_{Ax} = 25 \cos 40$ $\quad \vec{F}_{Ay} =$

$\vec{F}_{gx} = 0$ N $\quad \vec{F}_{gy} = -$

$\sum F_x = 19.2$ N $\quad \sum F_y$

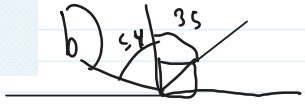
$\sum \vec{F} = \sqrt{19.2^2 + 11.8}$

$\sum \vec{F} = 22.6$ N

$\theta = \tan^{-1}\left(\frac{11.87}{19.2}\right) =$

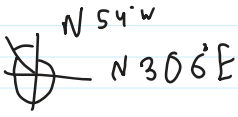
$\sum \vec{F} = 22.6$ N [31.7°

b)



$\vec{F}_{A1x} = 15 \sin 35$ $\quad \vec{F}_{A1y}$

$\vec{F}_{A2x} = 25 \sin 30$ $\quad \vec{F}_{A1y} =$



$\sum \vec{F}_x = -11.6$ N $\quad \sum \vec{F}$

$\theta = \tan^{-1}\left(\frac{26.98}{-11.6}\right)$

$\theta = [N - 66.734 E]$ $\quad \sum \vec{F} = \sqrt{-11.6^2 +}$

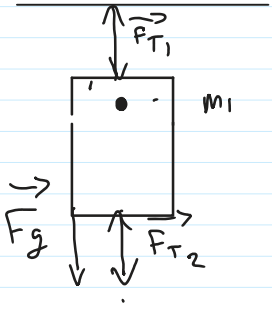
$\sum \vec{F} = 29.38$ [W 2

$\theta = [N\ 23.26\ W]$

3. **Figure 10** shows three masses connected by wires and hung vertically. Draw an FBD for each mass, and determine the tensions in the three wires. K/U T/I A [ans: top wire: 3.4×10^2 N; middle wire: 2.0×10^2 N; bottom wire: 1.3×10^2 N]

Figure 10

M_1)



$\vec{F_{T_1}} - \vec{F_{T_2}} = mg$ $\quad | \quad m = 15$

$= (20 + 15)\ 9.8$

$= 343$ N

- 25 sin 40

4.2 N

= 11.81 N

$\overline{7^2}$

01.7 25° above the horizon

]

= 15 cos 35

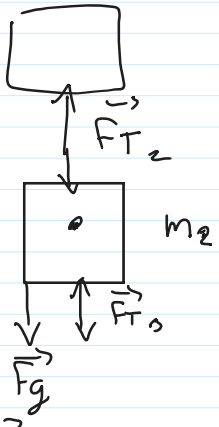
25 cos 30 6

y = 26.98 N

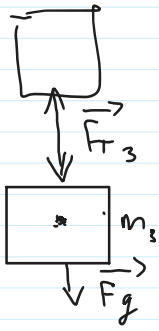
$\overline{26.98^2}$

3.26 W]

M2)



M3)



$\vec{F_{T_2}} = mg$

$= (13 + 7)(9.8)$

$= 196\ N$

$\vec{F_{T_3}} = mg$

$\vec{F_{T_3}} = 13 \times 9.8$

$\vec{F_{T_3}} = 127.4\ N$

9. You exert a force of 55 N on a heavy sled as shown in **Figure 12**. The force of gravity acting on the sled is 120 N [down]. The sled does not move across the rough horizontal surface, and the net force is zero. T/I

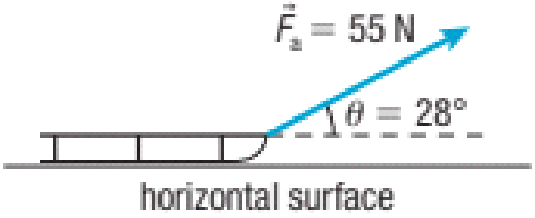
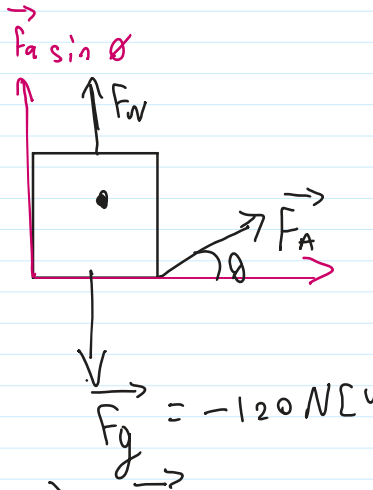


Figure 12

(a) Draw an FBD of the sled.

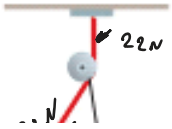
(b) Determine the normal force acting on the sled. Why is the magnitude of the normal force less than the magnitude of the force of gravity? Explain your answer.

(c) Calculate the force of static friction acting on the sled.

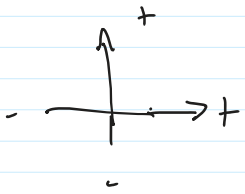


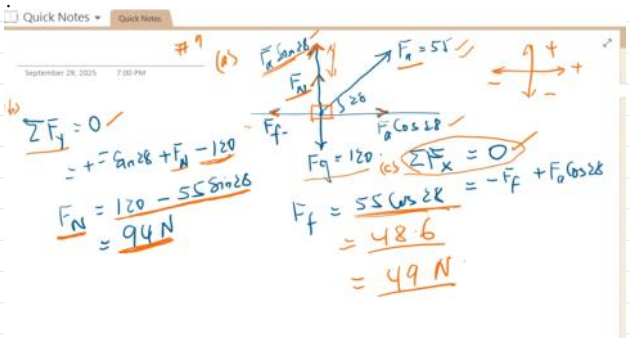
b) $\vec{F}_{Ay} = F_A$
$= 55$
$\vec{F}_{Ay} = 25$

2. Study the traction system shown in **Figure 11**. The tension in the vertical cord above the mass is 22 N. A student claims that the tension in the vertical cord above the leg must be more than 22 N. Discuss the validity of this statement. K/U T/I C



2) The tension is the complete string.





$\vec{F_A} \cos \theta$

$\text{up}] \quad \Sigma \vec{F} = 0$

$\sin \theta$

$\sin 28$

$.821 \text{ N [up]}$

Same all throughout, as it is 1

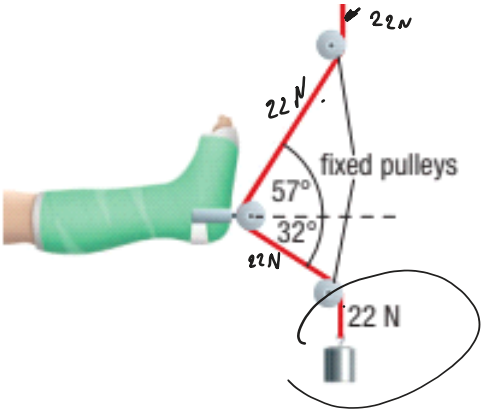


Figure 11

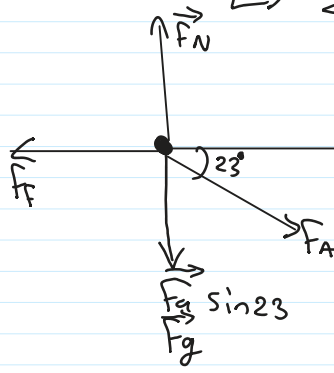
3) If y

3. Explain why ropes can only pull and never push. K/U

4) constant

4. You push your textbook at a constant velocity to the right across the table by applying a force at an angle of 23° below the horizontal. K/U T/I C A
   (a) List the forces acting on the textbook.
   (b) Draw an FBD of the textbook.

b)



a) F_N, F_f, F_g, $F_A \sin 23$, $F_A \cos 23$.

5)

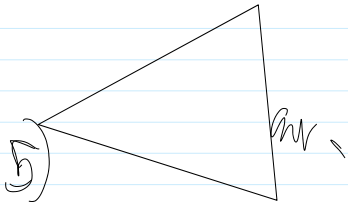
ou push on a rope, it will sag.

$\vec{V} = 0\ \vec{a}$

$\Sigma \vec{F_x}$ and $\Sigma \vec{F_y} = 0$

$\longrightarrow \vec{F_A} \cos 23$

5)



Questions

1. Summarize the common forces in a table with the following headings: Name, Symbol, Contact/Non-contact, Direction, Example in daily life. K/U C
2. Study the traction system shown in **Figure 11**. The tension in the vertical cord above the mass is 22 N. A student claims that the tension in the vertical cord above the leg must be more than 22 N. Discuss the validity of this statement. K/U T/I C

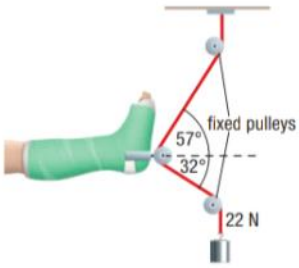


Figure 11

3. Explain why ropes can only pull and never push. K/U
4. You push your textbook at a constant velocity to the right across the table by applying a force at an angle of 23° below the horizontal. K/U T/I C A
   (a) List the forces acting on the textbook.
   (b) Draw an FBD of the textbook.
5. Given the forces $\vec{F}_A = 2.3$ N [S 35° W], $\vec{F}_B = 3.6$ N [N 14° W], and $\vec{F}_C = 4.2$ N [S 24° E], calculate the following:
   (a) $\vec{F}_A + \vec{F}_B + \vec{F}_C$
   (b) $\vec{F}_B - \vec{F}_C$ T/I
6. Given $\vec{F}_A = 33$ N [E 22° N] and $\vec{F}_B = 42$ N [S 15° E], calculate the force $\vec{F}_C$ needed so that $\vec{F}_A + \vec{F}_B + \vec{F}_C$ is zero. T/I
7. At the beach, three children pull on a floating toy. Child 1 pulls with a force of 15 N [N 24° E], child 2 pulls south, and child 3 pulls west. The net force on the toy is zero. Assume that there are no other significant forces acting on the toy. T/I
   (a) Calculate the magnitude of the forces exerted by child 2 and child 3 on the toy.
   (b) Child 2 lets go, and the other children maintain the same force. Calculate the net force on the toy.
   (c) What force must child 3 exert on the toy to cancel the force of child 1 on her own?
8. During a competition for charity, two students push horizontally on a heavy cart during a race. The net force on the cart is 180 N [E]. One student pushes with a force of 120 N [E 14° S]. Calculate the force that the second student exerts on the cart. T/I
9. You exert a force of 55 N on a heavy sled as shown in **Figure 12**. The force of gravity acting on the sled is 120 N [down]. The sled does not move across the rough horizontal surface, and the net force is zero. T/I

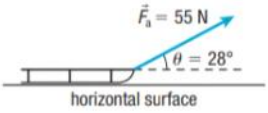
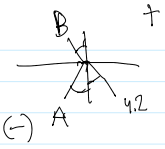


Figure 12

(a) Draw an FBD of the sled.
(b) Determine the normal force acting on the sled. Why is the magnitude of the normal force less than the magnitude of the force of gravity? Explain your answer.
(c) Calculate the force of static friction acting on the sled.

5) $F_a = 33$ [E 22 N
$F_B = 42$ [S 15 E]
$F_c = ?$
$F_a + F_B + F_c = 0$
$-F_c = F_A + F_B$
$F_{AB,x} = 33 \cos 22$
$F_{AB,y} = 33 \sin 22$
$F_{AB} = \sqrt{44.46}$
$F_{AB} = 50.151$
$-F_c = 50.151$
$F_c = -50.151$
$\theta = \tan^{-1}$
$\theta = 34.2$
[E
$F_c = -50.$



5) $F_A = 2.3$ N [S 35 W]
$F_B = 3.6$ N [N 14 W]
$F_c = 4.2$ N [S 24 E]

a) $F_A + F_B + F_c = F_T$

$F_{Tx} = -2.3 \sin 35 + -3.6 \sin 14 + 4.2 \sin 24 = -0.981$

$F_{Ty} = -2.3 \cos 35 + 3.6 \cos 14 + -4.2 \cos 24$

$F_T = \sqrt{(-0.981)^2 + (-2.2278)^2}$

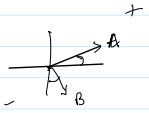
$F_T = 2.279$ N

b) $F_B - F_c$

F_{Tx}: $-3.6 \sin 14 - 4.2 \sin 24$

F_{Ty}: $3.6 \cos 14 - -4.2 \cos 24$

]

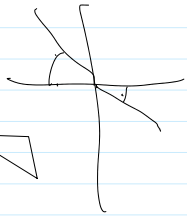


$+ 42 \sin 15 = 41.467$

$2 - 42 \cos 15 = -28.206$

$\sqrt{\quad ^2 + (-28.206)^2}$

1



$\left(\frac{28.206}{41.468}\right)$

23

$34.25]$

$15 \quad [E 34.25]$

$F_T = \sqrt{-2.579^2 + 7.329^2}$

$F_T = 7.770$

7. At the beach, three children pull on a floating toy. Child 1 pulls with a force of 15 N [N 24° E], child 2 pulls south, and child 3 pulls west. The net force on the toy is zero. Assume that there are no other significant forces acting on the toy. T/I
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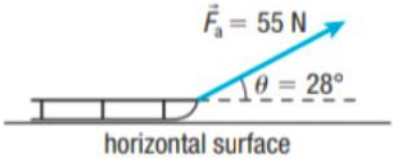
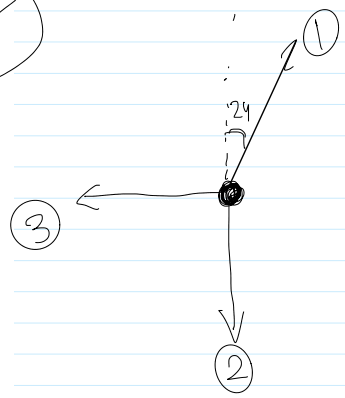


Figure 12

   (a) Draw an FBD of the sled.
   (b) Determine the normal force acting on the sled. Why is the magnitude of the normal force less than the magnitude of the force of gravity? Explain your answer.
   (c) Calculate the force of static friction acting on the sled.

7)



$\Sigma F = 0$

a) 2 =

$\Sigma f_x = 0$

$\Sigma f_y = 0$

$\Sigma F_y = 15 \cos 24 + -f_s = 0$

$\Sigma F_x = 15 \sin 24 + -f_w =$

$f_w = 6.10105$ [W]

$f_s = 13.703$ [S]

c) 15 N [S24W]

8) $\Sigma f = 180$ N [E]

$f_1 = 120$ N [E 14 S]

$f_2 = ?$

$f_{1x} + f_{2x} = f_x$

$f_{1y} + f_{2y} = f_y$

b) $15\,[N\ 24E] + 13.703\,[W] + \cancel{6.101\,[S]}$

$\Sigma F_y = 15\cos 24 - 0$

$\Sigma F_x = 15\sin 24 + - 6.101 = 0$

0 $\Sigma F = \sqrt{13.703^2}$

$\Sigma F = 13.703\ [N]$